

Nd:YAG Laser

**NEODYMIUM : YTTRIUM
ALUMINIUM GARNET LASER**

Nd: YAG Laser

- It is a solid state four level laser
- The **neodymium** (Nd^{+3}) ion is a rare earth metallic ion.
- Nd-YAG laser consists of Nd^{+3} ions doped (as impurity) in YAG crystal.
- The maximum concentration of Nd^{+3} ions in $\text{Y}_3\text{Al}_5\text{O}_{12}$ (**Yttrium aluminium garnet**) is about 1 to 1.5% (The increase concentration results in decrease of lifetime of metastable state.)

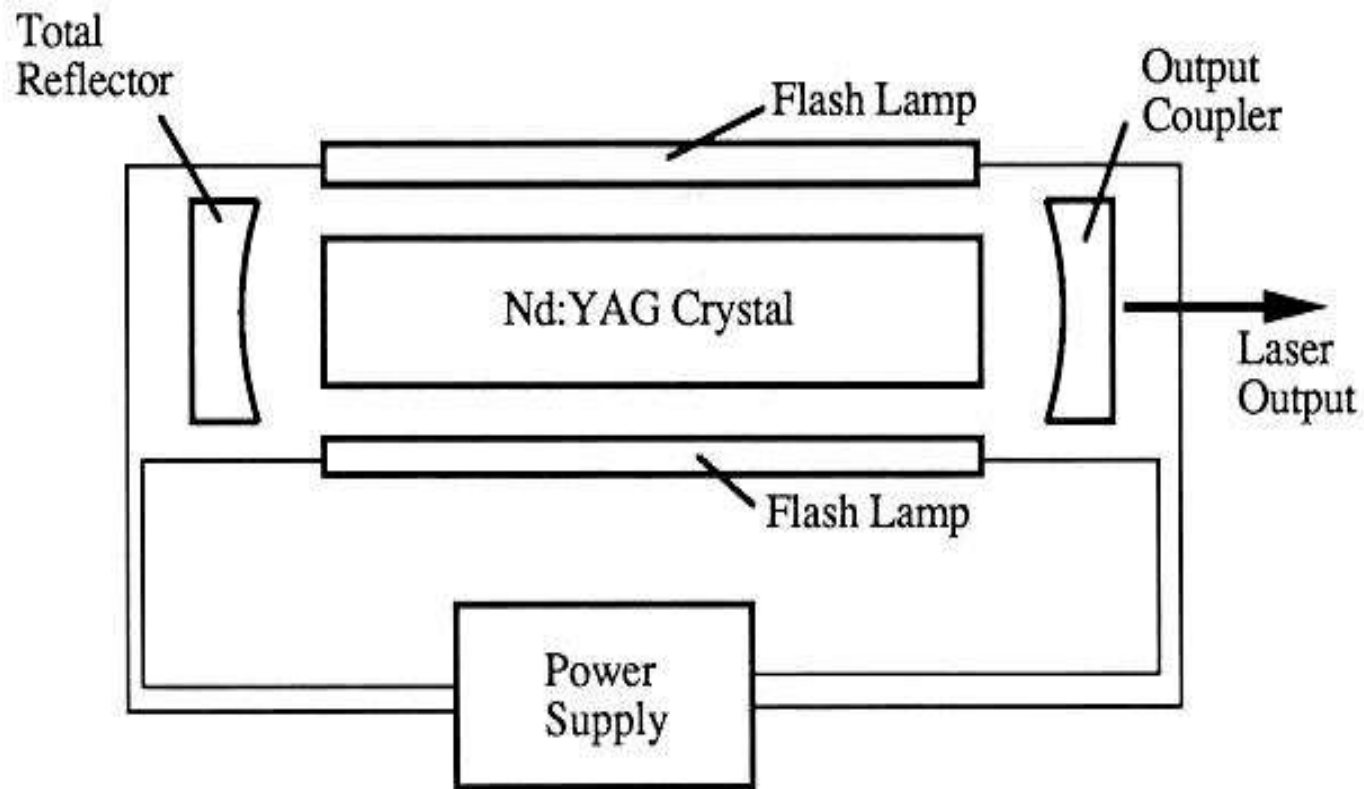
Contd..

- The YAG crystal is transparent and colorless. When doped with approximately 1% Nd, the crystal takes on a light blue color.
- Nd^{+3} ions occupy Yttrium ion sites & provide lasing transitions
- Role of YAG: (a) Nd^{+3} ions subjected to crystal field of host crystal YAG leads to splitting of energy levels of Nd^{+3} ions.
(b) Transition probabilities of various energy levels of Nd^{+3} ions are modified so that some transitions which are forbidden in free ion become allowed.

Construction :

- Essential parts of laser:-
 - (i) Yttrium aluminium garnet in which 1.5% of neodymium ions are doped as impurities. Nd^{+3} is the lasing ion.
 - (ii) Two end mirrors M_1 & M_2
 - (iii) Excitation is achieved by krypton or xenon **flash lamps**

Generally, a cooling system is required for operation of ND:YAG lasers. With an efficiency of about 3%, a typical ND:YAG produces thirty times as much waste heat as laser output; this heat must be removed in order to ensure proper laser operation.



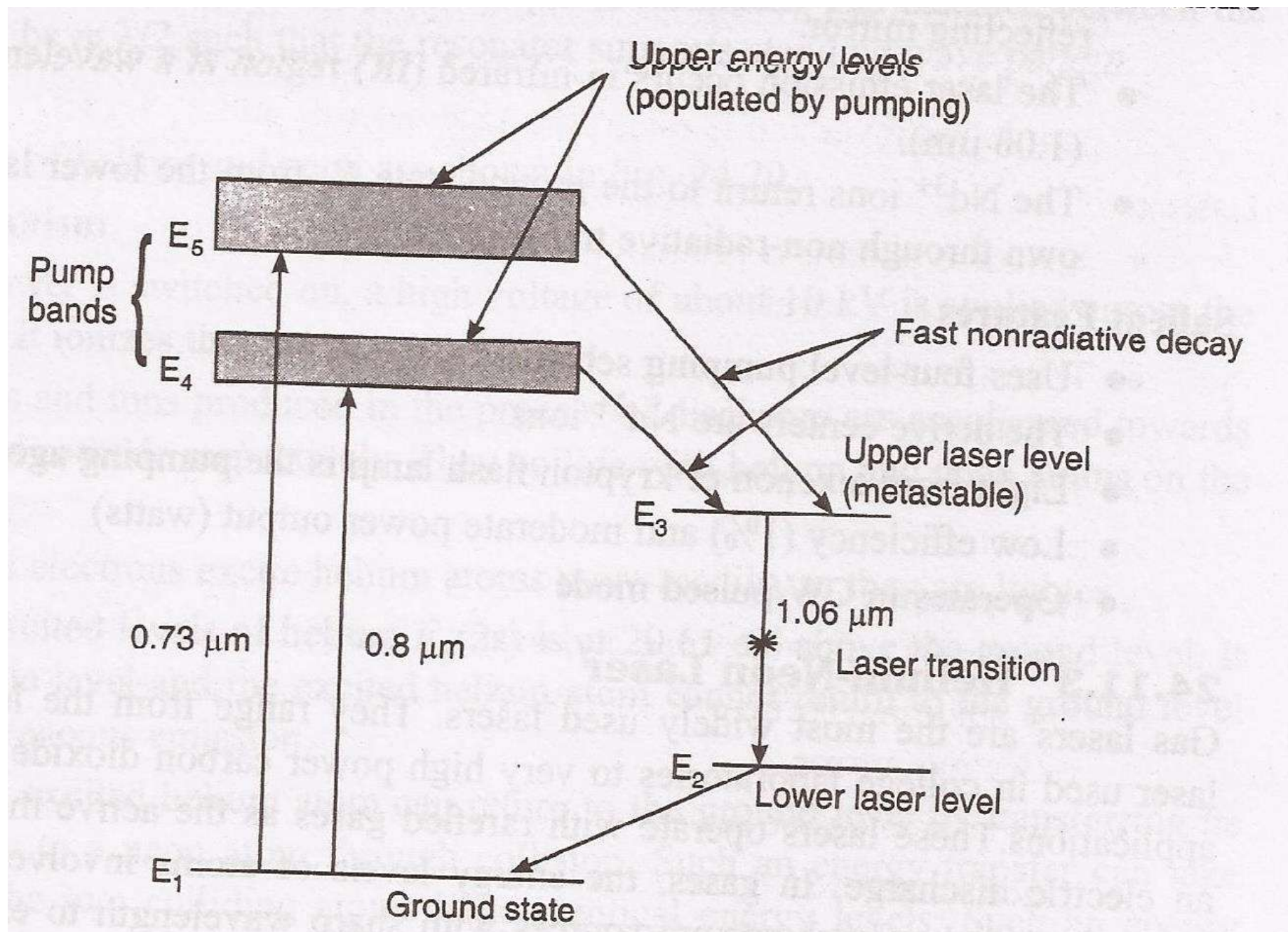
Nd : YAG laser rod has a length of about 10 cm with a diameter 12mm. It is kept at one foci of an elliptical glass tube.

Krypton flash lamp , the optical source is placed at the other foci of the glass tube. It is provided with necessary power supply arrangement.

Mirrors M_1 (total reflector)& M_2 (output coupler) act as a resonant cavity to produce stimulated and amplification process.

WORKING:

- When flash lamp is energised it gives out radiations
- Nd^{+3} ions get excited to higher energy levels by absorbing $0.73\text{ }\mu\text{m}$ and $0.80\mu\text{m}$ from the input radiations.(white light)
- Nd^{+3} ions can stay for a duration of about 10^{-8} sec.
- These ions undergo non-radiative transitions to reach the metastable state E_3 .
- Laser output of wavelength $1.06\text{ }\mu\text{m}$ is obtained from E_3 to E_2 transition.



Energy Levels of Neodymium ions in a crystal

Applications

- Material processing, such as [drilling](#), [cutting](#) and [welding](#) steel and super alloys.
- Medical applications([hair removal](#), and treatment of minor vascular defects).
- For [soft tissue surgeries](#) in the [oral cavity](#)
- In manufacturing as a means for engraving, etching, or marking a variety of metals and plastics.
- the most common laser used in laser designators and laser rangefinders.